Truth-table CEC

Truth-table CEC enumerates every PI assignment , fine for tiny circuits, infeasible for wide buses

The idea

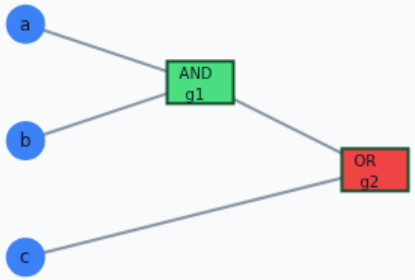
- Truth_table_cec walks all 8 rows; returns equivalent=True or counterexample dict

Eight rows

TRUTH-TABLE CEC

Eight rows

Step 1 / 5 · eight · module02-01-truth-table-cec



Explanation

Three PIs → 8 assignments to exhaust.

- 2^n

```
rows=8
```

Eight rows

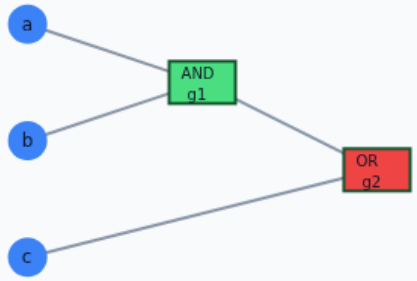
Good passes

TRUTH-TABLE CEC

Good passes

Step 2 / 5 · good · module02-01-truth-table-cec

{"equivalent":true,"rowsChecked":8,"counterexample":null}



Explanation

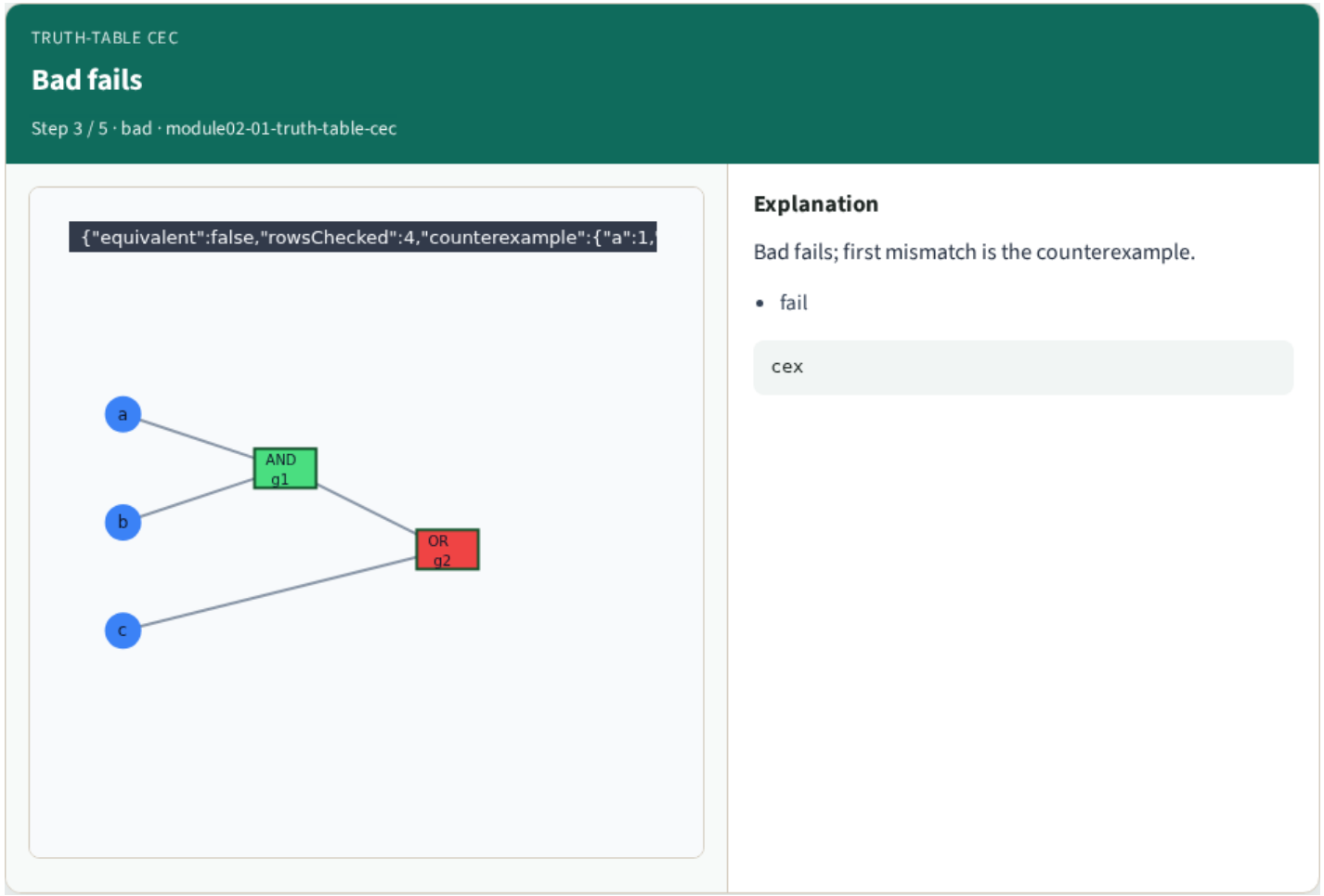
truth_table_cec(spec, good) → equivalent true.

- pass

```
8 checked
```

Good passes

Bad fails



Bad fails

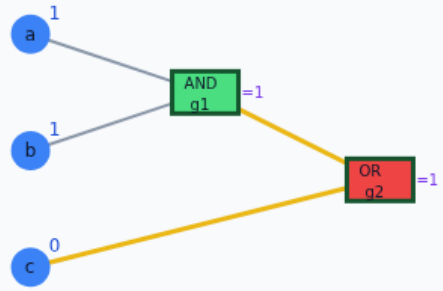
Cex a1b1c0

TRUTH-TABLE CEC

Cex a1b1c0

Step 4 / 5 · cex · module02-01-truth-table-cec

```
cex {"a":1,"b":1,"c":0}
```



Explanation

GOLDEN counterexample a=1, b=1, c=0.

- witness

```
spec1 bad0
```

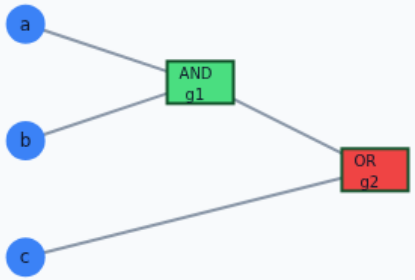
Cex a1b1c0

Scales poorly

TRUTH-TABLE CEC

Scales poorly

Step 5 / 5 · scale · module02-01-truth-table-cec



Explanation

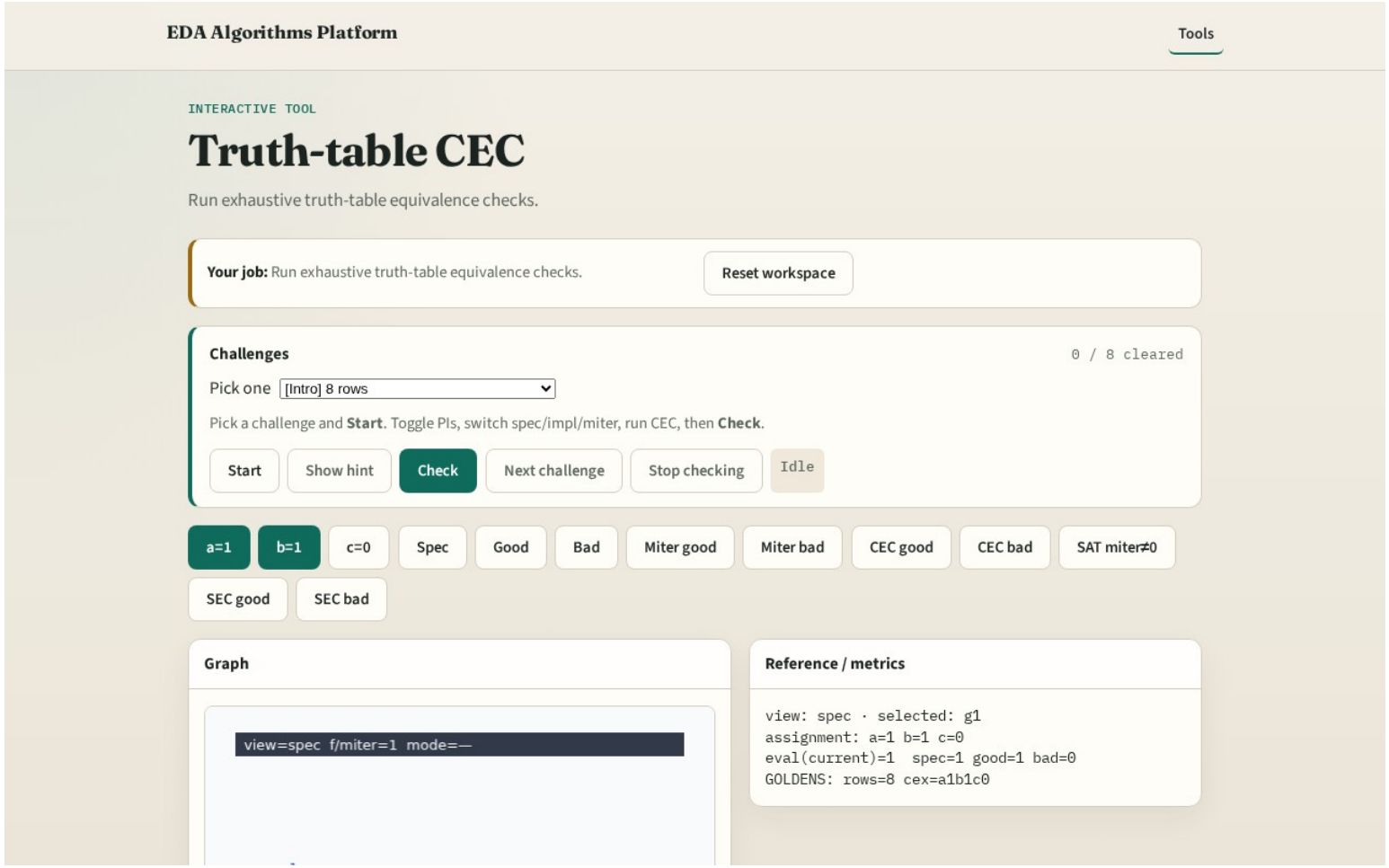
Exhaustive CEC is literacy; real tools use SAT/BDD.

- toy only

```
Next hash
```

Scales poorly

Browser lab track



Browser lab starter

Implement track

- Call truth_table_cec(spec, impl_good) and (spec, impl_bad); assert goldens

Pitfalls

- Stopping at first match instead of first mismatch
- Missing a PI bit in enumeration

Your turn

- Next: structural hashing / FRAIG lite